

BEYOND NODAL DAMPING

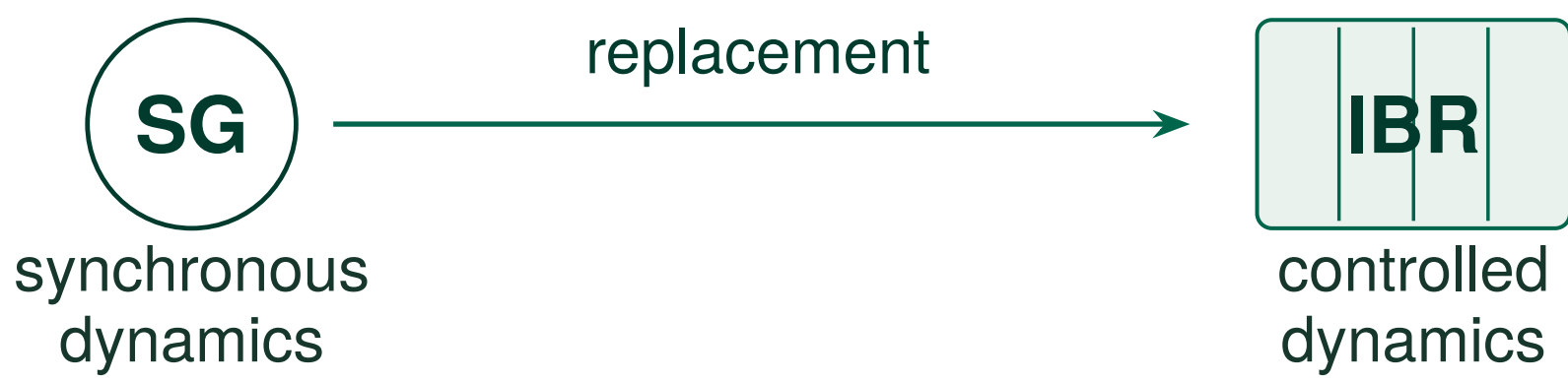
WHEN STABLE REPLACEMENTS FAIL TOGETHER

Minimal Incompatibility Hypergraphs for Safe SG → IBR Transition Planning

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1 MOTIVATION & QUESTION

Replacing SG units changes the full portfolio. A stable endpoint can still contain an unstable intermediate coalition.

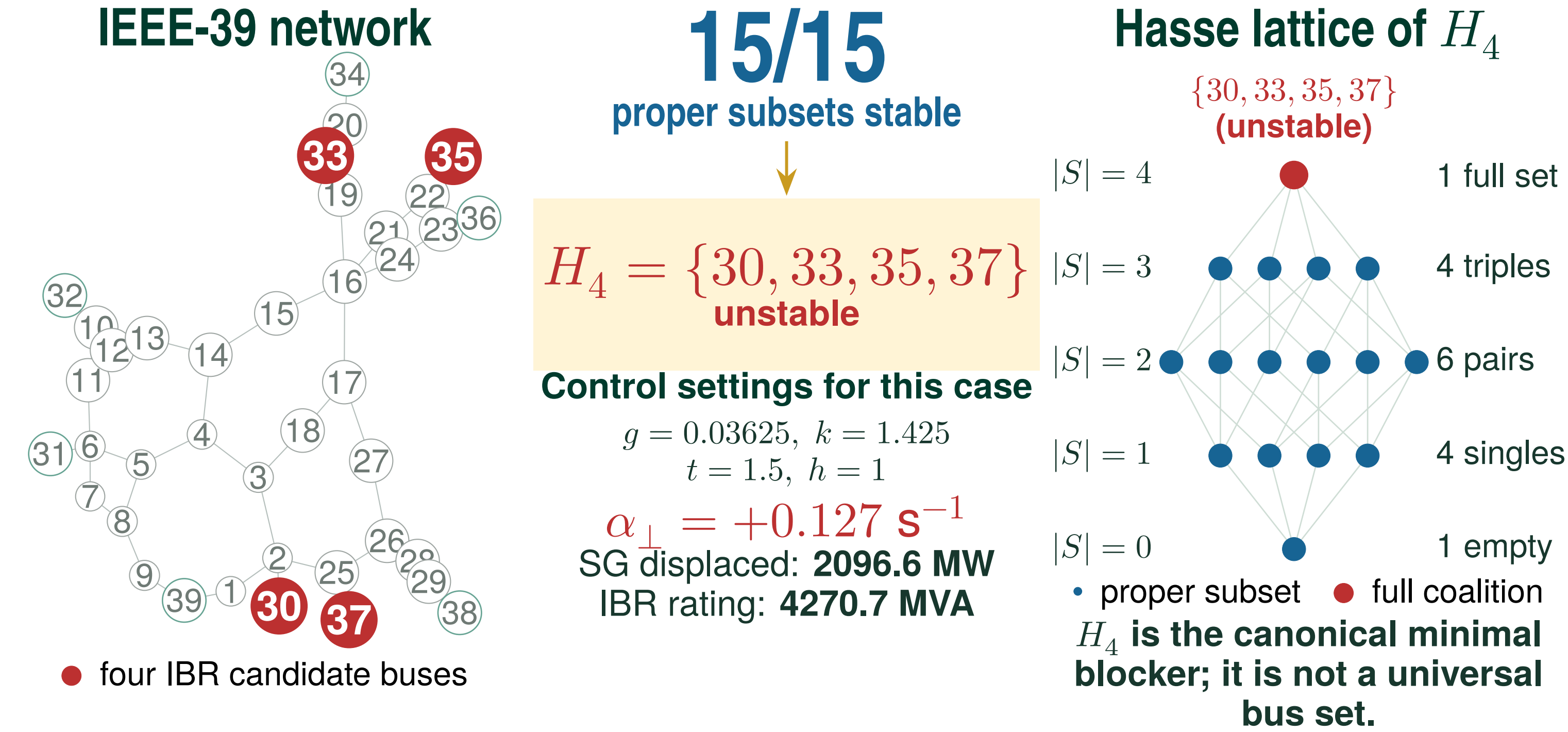


Which replacement combinations are incompatible, why do they fail, and how can paths avoid them?

$$\mathcal{H}(\theta) = \text{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Edges mark minimal unstable coalitions.

2 FLAGSHIP RESULT: IEEE-39



3 STABLE TARGET $\neq$ SAFE PATH

Three nested guarantees for a target portfolio T :

A FINAL STATE STABLE $\alpha_{\perp}(T, \theta) < 0$	B SOME ORDER IS SAFE Stable prefix chain exists	C EVERY ORDER SAFE No blocker in T .
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$C \Rightarrow B \Rightarrow A$

Converse implications can fail.

$$C(T) \iff \forall e \in \mathcal{H}(\theta), e \not\subseteq T$$

Recovery of exact blocker sets

Matches across 52 operating points

first-order modal 22/52

additive 31/52

pairwise 38/52

15/15
smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

$+0.127 \text{ s}^{-1}$
critical growth rate
canonical portfolio

3/327
order-dependent targets
matched-policy IEEE-39 census

4A FROM DEVICES TO NETWORK RETURN

Network update $T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$

Characteristic $\det P_S(s) = h_S(s) \det T_0(s) \det(I + M_{SS}(s))$

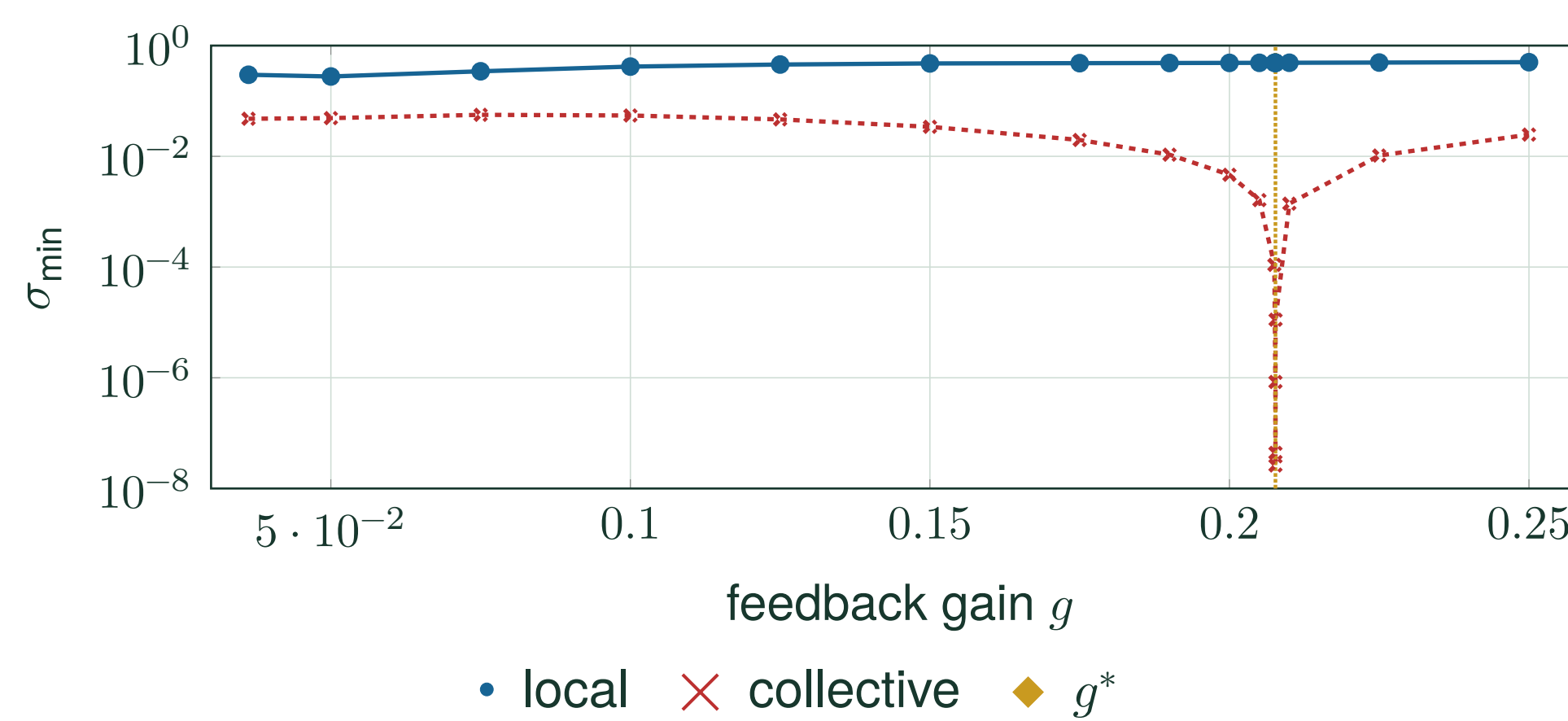
Local $\times$ collective $\det(I + M_{SS}) = \left[\prod_{i \in S} \det(I + M_{ii}) \right] \det(I + Q_{SS})$

Interactions $\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$

Full-order factorization; no reduced-mode equivalence is claimed.

4B THE COLLECTIVE TERM CLOSES

Boundary scan: $g^* = 0.207681$; $\alpha_{\perp} = 0$; $f = 0.7064 \text{ Hz}$



LOCAL DEVICES

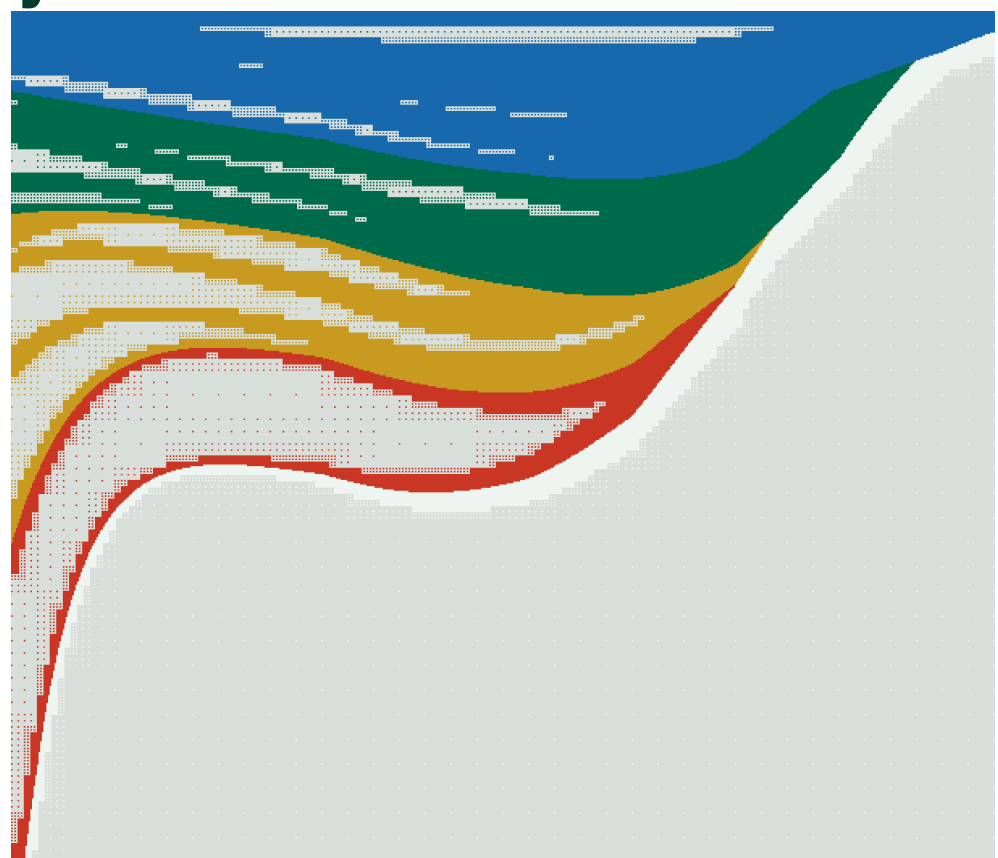
$\sigma_{\min} \approx 0.49$
remain regular

NETWORK RETURN

$\sigma_{\min} \approx 4.4 \times 10^{-8}$
approaches singularity

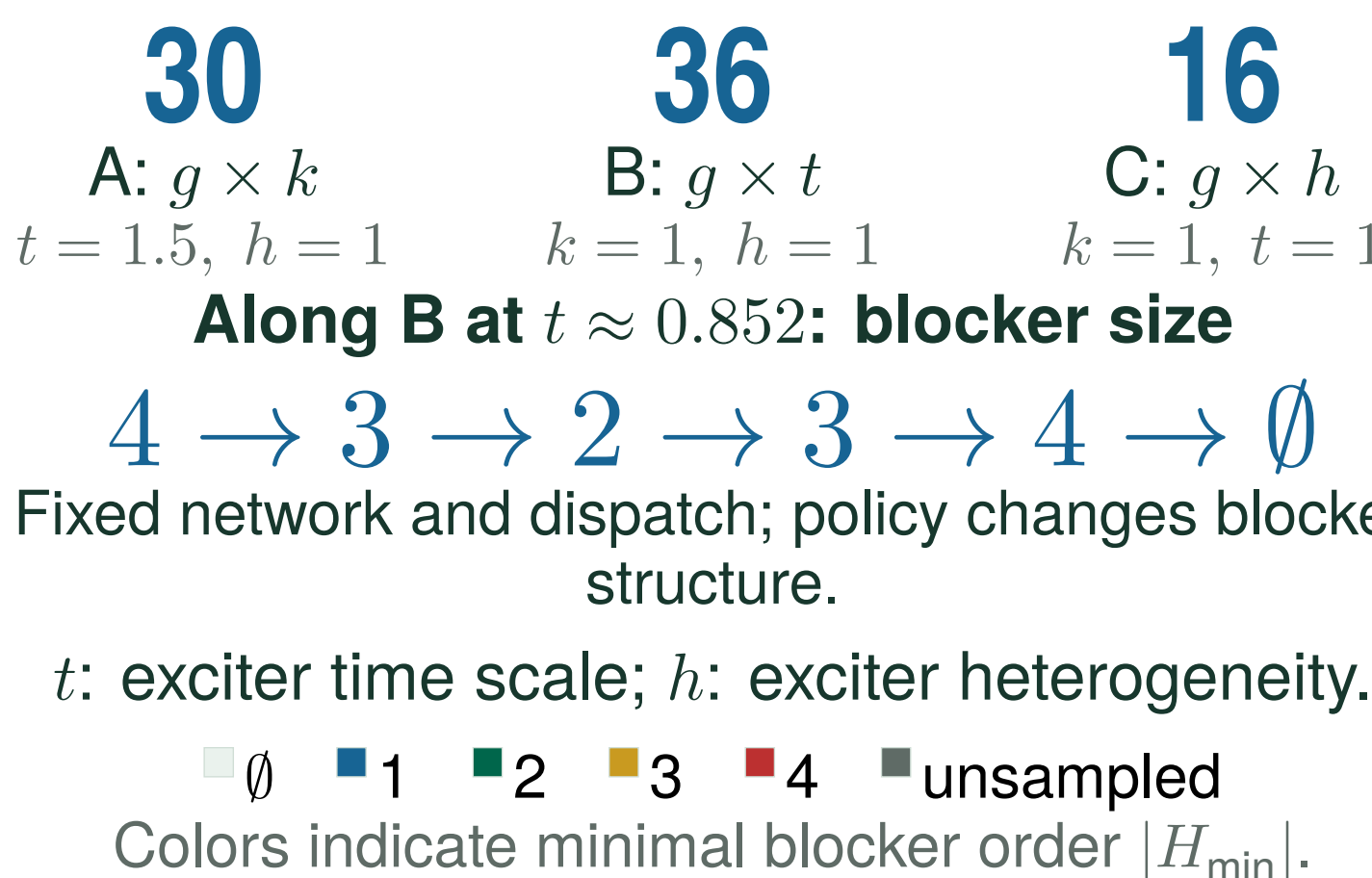
5 POLICY CHANGES THE MINIMAL BLOCKER

Policy atlas A: four IBR candidates



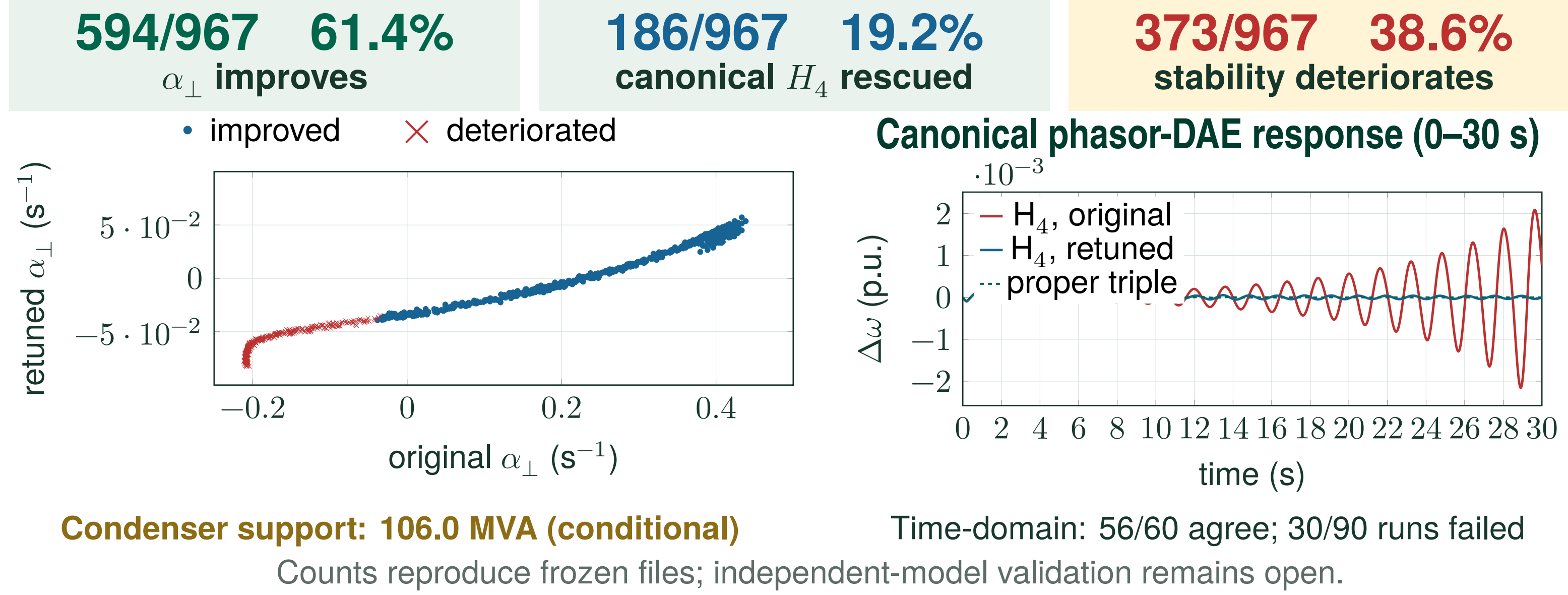
Q/V gain g (horizontal); AVR gain k (vertical).

Distinct full-spectrum hypergraphs



7 A CANONICAL REPAIR IS NOT UNIVERSAL

Synthetic retuning: $g = 0.03625 \rightarrow 0.25$

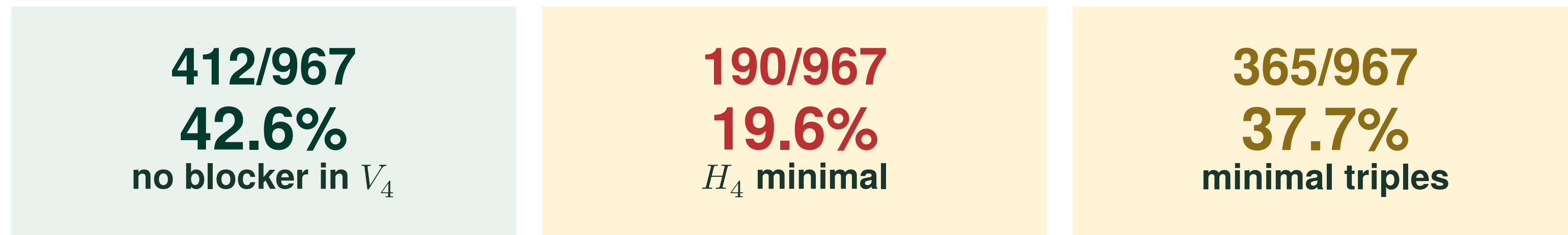


6 THE CANONICAL H_4 IS NOT UNIVERSAL

Frozen synthetic envelope (1000 scenario IDs)



33 infeasible (3.3%) Fractions below use 967 valid IDs.



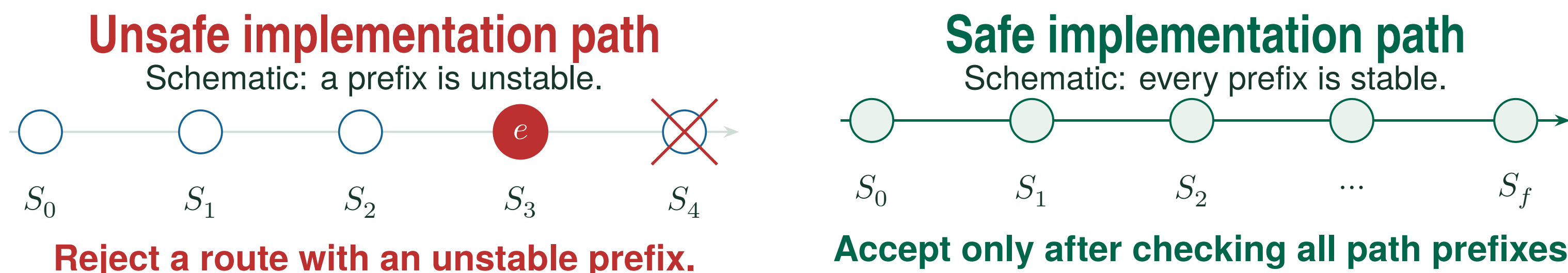
Partition of valid cases; 555/967 (57.4%) have a collective blocker.
Frozen synthetic ensemble; descriptive counts, not population probabilities.

8 SAFE PATHS CHECK EVERY PREFIX

Path-constrained planning formulation

$$\max \sum_{i \in S_f} P_i \quad \text{s.t. } S_0 = \emptyset, S_{t+1} = S_t \cup \{i_t\}, \alpha_{\perp}(S_t, \theta) < 0 \quad \forall t$$

Matched-policy IEEE-39 census: 3/327 stable targets depend on order.



The order matters: test stability after each replacement.

9A MODEL CHECKS



9B WHAT “SAFE PATH” MEANS HERE

Each prefix is a re-equilibrated phasor-DAE portfolio.
The analysis does not simulate physical switching, EMT, current limits or DC-link limits.
Blockers depend on the model and control policy.

TAKEAWAY

SAFE PREFIXES, SAFE TRANSITIONS.

A stable endpoint can still hide an unstable intermediate coalition.
Blockers change with the control policy.
Safety here refers to re-equilibrated phasor-DAE prefixes.

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REFERENCES

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